Labwork 3

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1 Network Architecture Design and Implementation

We design the neural network using three basic parts: `Neuron`, `Layer`, and `NeuralNetwork`. This design helps us easily change the size of the network.

1.1 Main Components

- **Neuron Class:** This is the smallest part. Every neuron saves its own weights (w) and a bias (b). It also uses the Sigmoid function to calculate the output.
- **Layer Class:** This is a collection of neurons. When we create a layer, it makes many neurons. All neurons in this layer get the same inputs from the previous layer.
- **NeuralNetwork Class:** This is the main controller. It reads the network structure from a file named `struc.txt` and creates the layers. It also sets the weights and biases from `weight.txt`.

1.2 How the Network is Built

The network is built in two clear steps:

1. **Read the Structure:** The network opens `struc.txt`. The first line shows the total number of layers. The next lines show how many neurons are in each layer.
2. **Create Layers:** The system loops through the structure list. It calculates the input size and the number of neurons for each layer, then creates them one by one.

2 Feedforward Implementation

The feedforward process moves data forward through the network. It goes from the input layer, through the hidden layers, and finally to the output layer.

2.1 Math Formulas

Inside each neuron, we multiply the inputs by their weights, add them together, and add the bias. Then, we use the Sigmoid function to get a result between 0 and 1.

$$z = b + \sum_{i=1}^{n} w_i \cdot x_i \tag{1}$$

$$\text{output} = \sigma(z) = \frac{1}{1 + e^{-z}} \tag{2}$$

2.2 Step-by-Step Execution

- **Network Step:** The `feedforward` method gets the first input data. It uses a loop to pass this data through every layer. The output of one layer becomes the input for the next layer.
- **Layer Step:** The `forward` method in the Layer class sends the input data to every neuron inside that layer. It collects all the neuron results into a list.
- **Neuron Step:** The `forward` method in the Neuron class does the math. It calculates the sum, adds the bias, and calls the sigmoid function to return the final number.